

ALI HASSAN

AI/ML Engineer | Generative AI & Multi-Agent Systems

Lahore, Pakistan | 0307 1532386 | 0305 4751551 | alihassanu066@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production AI agents, multi-agent systems, and LLM-powered applications using LangChain, LangGraph, and CrewAI. Built QA automation and token monitoring systems to improve agent reliability and control LLM costs. Develops scalable backends with FastAPI and Supabase, with working knowledge of SQL and Python data processing.

TECHNICAL SKILLS

Expertise: Generative AI, AI Agents, Multi-Agent Systems, RAG, Prompt Engineering, LLM Evaluation & Token Monitoring, Machine Learning

Tools & Technologies: Python, LangChain, LangGraph, CrewAI, Gemini, Claude, GPT, Pinecone, ChromaDB, pgvector, PyTorch, scikit-learn, FastAPI, PostgreSQL, Supabase, Redis, Docker, Git/GitHub, Vitest, Playwright, pandas, SQL

PROFESSIONAL EXPERIENCE

Junior AI Engineer — Friday Media Group

January 2026 – Present

Built a QA Supervisor Agent that automatically discovers, tests, and monitors production AI agents across Finance, Marketing, and Customer Support, using hybrid evaluation (rule-based checks + LLM-as-judge) and regression testing against saved baselines.

- Built a zero-impact instrumentation shim across 142 edge functions that simulates and records side effects (DB writes, emails, refunds, API calls), enabling safe dry-run testing without affecting production data.
- Implemented per-test LLM token and cost tracking (input/output tokens, model calls, separate judge costs), giving clear visibility into the cost of testing each agent.
- Developed 700+ automated Vitest/Playwright tests across 35 test categories, covering functionality, security, prompt injection, concurrency, reliability, and edge cases.
- Built a real-time QA dashboard with Supabase Realtime for execution traces, token usage, and regression tracking.

AI Engineering Intern — Friday Media Group

October 2025 – December 2025

- Built multi-agent systems with CrewAI and LangChain, orchestrating specialized agents for complex tasks.
- Developed RAG pipelines using Pinecone and ChromaDB to deliver context-aware AI responses.
- Built scalable FastAPI services backed by Supabase/PostgreSQL and Redis, with caching, session management, and query optimization.
- Created Python data pipelines (Pandas, NumPy) for data processing and transformation, following SOLID principles and clean architecture.

PROJECTS

Adalat – AI Courtroom Simulator

Python • Gemini • Streamlit • Pydantic • SQLite

- Built a multi-agent legal reasoning simulator where prosecutor, defence, and judge agents argue cases under Pakistani law (PPC, CrPC, QSO, PECA), with a judge that delivers structured verdicts against a weighted rubric.
- Designed hybrid retrieval (BM25, citation matching, HyDE) fused with reciprocal rank fusion, achieving 100% recall@5 on a 40-case benchmark, plus a 3-layer defence against hallucinated law (retrieval-first prompting, ID stripping, LLM verifier).

Awaaz – Real-time Voice AI Assistant

LiveKit Agents • ElevenLabs • FastAPI • pgvector • React

- Built a real-time bilingual (Urdu/English) voice assistant with a FastAPI backend, PostgreSQL + pgvector memory, Redis, and a React web client, fully containerized with Docker.

GoGuide – AI Travel Planning System

FastAPI • CrewAI • LangChain • Render

- Built and deployed a multi-agent travel planner where specialized agents (Location, Guide, Planner) generate personalized itineraries from destination, dates, budget, and interests.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

2022 – 2026

Lahore Leads University, Lahore

CERTIFICATIONS

- HCCDA-AI (Huawei Cloud Certified Developer Associate – AI) — NAVTTC (Sep 2025 – Nov 2025)
- Microsoft Dynamics 365 — NAVTTC (Feb 2024 – May 2024)
- Diploma in Computer Management — Police Computer Training School

LANGUAGES

English (Professional Working Proficiency) • Urdu (Native) • Punjabi (Native)